

Exercise 1

The data contained in the `cement` dataset represent the hardness (hardness variable) of 13 types of cement with different chemical compositions. Specifically, each type is obtained with varying proportions of aluminium (aluminium variable), silicate (silicate variable), calcium aluminoferrite (aluminium_ferrite), and silicate bic (silicate_bic). The interest is explaining how the hardness of cement depends on the proportions of chemicals.

A regression model was fitted for this purpose and produced the following result:

		Estimate	Std. Error	t statistic	Pr(> t)
β_0	(Intercept)	124.4809	26.7557	4.653	0.0016
β_1	aluminium	0.9739	??	3.435	0.0089
β_2	silicate	-0.1405	0.2891	-0.486	0.6400
β_3	aluminium_ferrite	-0.4974	0.2751	??	??
β_4	silicate_bic	??	0.3214	-2.481	0.0381

Error sum of squares

Total sum of squares

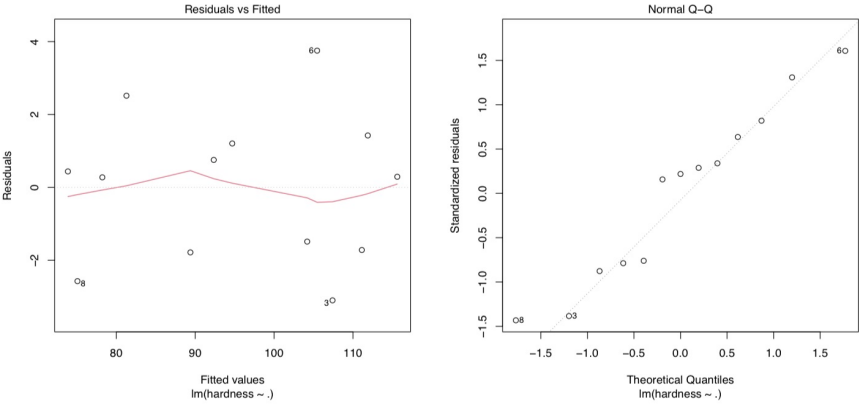
R^2 coefficient

49.378

2715.763

??

- a) Write the model formulation and assumptions.
- b) Complete the missing values in the table. For “Pr(> |t|)” of `aluminium_ferrite` provide an approximate value. What variables have a statistically significant effect?
- c) Test the statistical hypothesis corresponding to the statement “the covariates do not have an effect on the hardness of cement”.
- d) On a reduced model (“model B”) that includes only the variables `aluminium` and `silicate_bic` the error sum of squares is equal to $SSE_B = 74.762$. Perform an F test to compare this model with the complete model (“model A”) that includes all the covariates. Interpret the result: which model would you prefer?
- e) Obtain the coefficient R^2 of model B. Instead of performing the test in point (d), could you have simply compared the coefficient R^2 of the two models? Why?
- f) Figure 1 shows two plots regarding the complete model (model A). Explain what they represent and interpret them.



3

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$$

$$\epsilon_s^{obs} \sim \mathcal{N}(\epsilon_{n-p})$$

$$\epsilon_2^{obs} = \frac{\hat{\beta}_2 - \beta_2}{SE(\hat{\beta}_2)} \rightarrow SE(\hat{\beta}_2) = \frac{\hat{\beta}_2}{\epsilon_2^{obs}} = 0.2335$$

4

$$\begin{cases} H_0: \beta_4 = 0 \\ H_1: \beta_4 \neq 0 \end{cases}$$

$$\epsilon_9^{obs} \sim \mathcal{N}(\epsilon_{n-p})$$

$$\epsilon_4^{obs} = \frac{\hat{\beta}_4 - \beta_4}{SE(\hat{\beta}_4)} = -1.308$$

5

$$\chi^{obs} = P(|\epsilon_{n-p}| \geq |\epsilon_4^{obs}|) = 2 \min \{ P(\epsilon_{n-p} < -\epsilon_4^{obs}); P(\epsilon_{n-p} > \epsilon_4^{obs}) \} \approx 2(1 - 0.94) = 0.12$$

$$n-p = 13 - 5 = 8$$

5c

COVARIATES DO NOT HAVE AN EFFECT ON HARDNESS OF CEMENT

$$\begin{cases} H_0: \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0 \\ H_1: \text{THO} \end{cases}$$

$$f_{obs} \sim f_{p-p_0; n-p}$$

$\downarrow$

5-1

$\downarrow$

13-5

$$f_{obs} = \frac{\frac{SST - SSE}{p - p_0}}{\frac{SSE}{n - p}} = \frac{\frac{2765.141 - 49.378}{5 - 1}}{\frac{49.378}{13 - 5}} = 109.9$$

$$\alpha^{obs} = P(f_{p-p_0; n-p} > f_{obs}) \approx 1 - 0.99980$$

Since $\alpha^{obs} < \alpha$ for $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$ I REJECT H_0 AT ALL CONFIDENCE LEVEL

2a

$$n = 13$$

$$Y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i} + \beta_4 x_{3i} + \beta_5 x_{4i} + \epsilon_i$$

- $\mathcal{Y}_{:1} = x_{:1}$ $\mathcal{Y}_{:p} = x_{:p}$
- FIXED AND NON-STOCHASTIC
- NOT ALL EQUAL
- $E(\epsilon_i) = 0 \quad \forall i = 0$
- $Cov(\epsilon_i; \epsilon_k) = 0$ FOR $i \neq k \rightarrow Cov(Y_i; Y_k) = 0$ FOR $i \neq k$
- $VAR(\epsilon_i) = \sigma^2 > 0$ FOR $i = k$
- $\epsilon_i \sim \mathcal{N}(0; \sigma^2)$
- THE VECTOR $x_j \in \mathbb{R}^n \quad j = 1, \dots, p$ ARE LINEARLY INDEP $\rightarrow Rank(x) = p$ (FULL RANK)

6

		Estimate	Std. Error	t statistic	Pr(> t)
β_1	(Intercept)	124.4809	26.7557	4.653	0.0016
β_2	aluminium	0.9739	3 ??	3.435	0.0089
β_3	silicate	-0.1405	0.2891	-0.486	0.6400
β_4	aluminium_ferrite	-0.4974	0.2751	4 ??	5 ??
β_5	silicate_bic	2 ??	0.3214	-2.481	0.0381

Error sum of squares

Total sum of squares

R^2 coefficient

49.378

2715.763

1 ??

7

$$SSE = 49.378$$

$$SSR = 2715.763$$

$$SST = SSE + SSR = 2765.141$$

$$R^2 = \frac{SSR}{SST} = 0.992$$

2

$$\begin{cases} H_0: \beta_2 = 0 \\ H_1: \beta_2 \neq 0 \end{cases}$$

$$\epsilon_s^{obs} \sim \mathcal{N}(\epsilon_{n-p})$$

$$\epsilon_s^{obs} = \frac{\hat{\beta}_2 - \beta_2}{SE(\hat{\beta}_2)} \rightarrow \hat{\beta}_2 = \epsilon_s^{obs} \cdot SE(\hat{\beta}_2) = -0.7974$$

VARIABLES WITH STATISTICALLY SIGNIFICANT EFFECT AT 10% $\rightarrow \alpha^{obs} < 0.10$

- INTERCEPT
- ALUMINIUM
- SILICATE_BIC

⑥

$$\text{Model A: } Y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{2i} + \beta_4 x_{3i} + \beta_5 x_{4i} + \epsilon_i \quad \text{SSE}_A = 49.379$$

$$\text{Model B: } Y_i = \beta_1 + \beta_2 x_{1i} + \beta_3 x_{4i} \quad \text{SSE}_B = 71.762$$

$$\text{SST} = 2715.763$$

$$\begin{cases} H_0: \beta_3 = \beta_4 = 0 \\ H_1: \text{THO} \end{cases}$$

$$f_{\text{obs}} \sim F_{p-p_0, n-p}$$

$\downarrow$ 2 $\downarrow$ 8

$$f_{\text{obs}} = \frac{\frac{\text{SSE}_B - \text{SSE}_A}{p - p_0}}{\frac{\text{SSE}_A}{n - p}} = 2.0563$$

$$\alpha^{\text{obs}} = P(|f_{p-p_0, n-p}| > |f_{\text{obs}}|) > 0.10$$

SINCE $\alpha^{\text{obs}} > \alpha$ FOR $\alpha = 0.1; \alpha = 0.05; \alpha = 0.01$ I DO NOT REJ. H_0 I PREFER THE MODEL WITH ONLY SILICATE- β_3 AND ALUMINUM

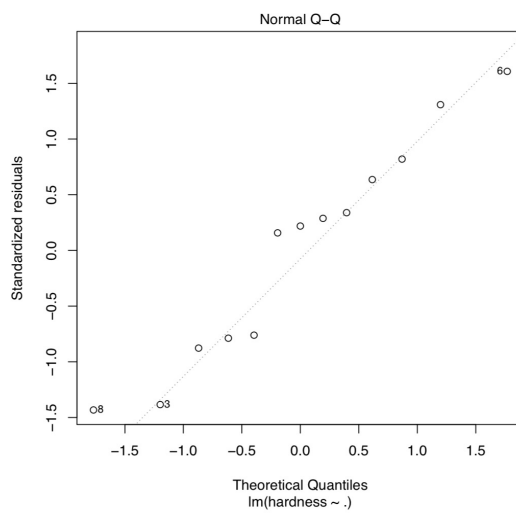
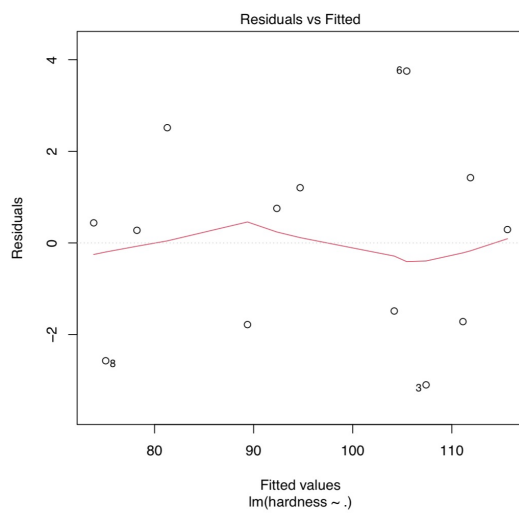
⑦

$$R_B^2 = \frac{\text{SSR}_B}{\text{SST}} = \frac{2641}{2715.763} = 0.972$$

$$\text{SSR}_B = \text{SST} - \text{SSE}_B = 2641$$

SINCE MODEL B HAVE LESS COVARIATES IT WILL HAVE HIGHER SSR AND SO A SMALLER R^2 (SINCE SST IS FIXED) AND SO IT IS OBVIOUS THAT R^2 OF B WILL BE SMALLER THAN R^2 OF ORIGINAL MODEL

⑧



- TOO FEW OBSERVATIONS TO MAKE ASSUMPTIONS
- CLOUD DISPERSION OF DOTS
- NO PATTERN OF RESIDUAL VISIBLE
- USE TO CHECK IF MODEL ASSUMPTION ARE SATISFIED (ORTHOGONALITY OF RESIDUALS AND HOMOCEASTITY)
- IF BAD PLOT (AND THERE ARE PATTERNS) WE NEED TO FIX IT
 - ADD OTHER COVARIATES
 - TRANSFORM COVARIATES

- TOO FEW OBSERVATIONS
- NO PATTERN IS VISIBLE
- SOME OUTLIERS IN THE TAIL
- NEED MORE OBSERVATIONS
- ERRORS FOLLOW NORMAL DISTRIBUTION
 - CHECK NORMALITY ASSUMPTION
- USE TO CHECK IF RESIDUALS OF THE REGRESSION MODEL ARE NORMALLY DISTRIBUTED
- IF ERROR FOLLOW A NORMAL DISTRIBUTION $\rightarrow$ POINTS ON THE LINE
- IF THERE IS A PATTERN
 - TRANSFORM COVARIATES
 - EXAMINE THE DATA
 - USE ANOTHER MODEL